

Do online psychological services start-ups have a big potential market in China

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I. DATA STORY SUMMARY

The data story is regarding the exploration of whether the online psychological services start-ups have a big potential market in China. The on-line psychological services contain the services:

- 1) Psychological counselling services
- 2) Psychotherapist training services
- 3) Training courses services for people who want to learn or broaden their psychological knowledge and improve psychological quality

To explore the topic, the questions can be summarized as follows:

1) What is the mental health situation in China? Does China have big psychological service requirements but lack mental health resources which can contain beds in varieties of mental health facilities, outpatient facilities, and mental health professionals. These resources are the most fundamental resources for severe psychiatric treatment, if a country lacks these resources, it can be inferred that the country more lacks the professional resources for common psychological disorder and psychological behaviour problems as well. Thus, obviously, the psychological services providing business will have a big potential market to meet customer's requirements.

2) Are China's mental health resources unevenly distributed? If yes, on-line psychological counselling services can use the highly developed internet technologies to solve the problem of uneven distribution of mental resources without considering geographical issues. This means all services can be processed via network call or video instead of face to face.

3) Does government's mental health spending per capita increase with Gross National Income rising? If yes, this indicates if a country's GNI keeps rising trend, the government will increase mental health spending investment, which means the business of providing psychological services will obtain the more government's support. Meanwhile, with the continuous increase in national income, the national consumption of mental health services will gradually enter an accelerating upgrade.

In terms of these questions, they will be answered in by means of these methods:

1) Use Bar Charts to compare China's mental health resources including beds in varieties mental health facilities, outpatient facilities, and mental health professionals with other countries' of WHO and the world average level to show the situation of China's mental

health resources in the world level whether they are below the world average level.

2) Use Choropleth Maps to show the distribution of psychiatric beds and medical professionals in China to indicate whether China's mental health resources are unevenly distributed.

3) Plot a scatter between the government's mental health spending per capita and Gross National Income and make a fitting model, to demonstrate the correlation between mental health spending per capita and Gross National Income and check whether mental health spending per capita increases with Gross National Income rising. Plot two lines chart of China's Gross National Income amount and per capita respectively over time to show the increasing trend.

II. DATASET SUMMARY

The datasets used in the data story can be summarised into 3 types in accordance with the previous categorisation of questions.

The first type datasets corresponding to the first question contain three datasets:

1) Beds in general hospitals for mental health, beds in mental hospitals and beds in community residential facilities (per 100,000 population) globally in 2011 and 2014 [1].

2) Outpatient facilities (per 100,000 population) globally in 2011 [2].

3) Psychiatrists and nurses (per 100,000 population) globally in 2011 and 2014 [3].

They are all from the website of WHO. Because in these datasets, China lacks some data, the data story has to pick the data which China has. The results are beds in general hospitals for mental health globally in 2011, beds in mental hospitals globally in 2011 and 2014, mental hospitals globally in 2011, and psychiatrists, psychologists and nurses globally in 2011.

The second type datasets corresponding to the second question are from two papers which present the corresponding datasets:

1) Psychiatric beds distribution of hospitals in different provinces of China in 2010 [4].

2) Mental health professionals working in 757 specialized mental health facilities in seven regions of China in 2010 [5].

The third type datasets corresponding to the third question include two datasets:

1) Mental Health Atlas-2014 country profiles describing the government's mental health spending per

capita of WHO member states, which come from the website of WHO [6].

2) The Gross National Income of WHO member states from 1992-2014 [7]. From the above two datasets, combine the government’s mental health spending per capita of countries which have data and their GNI.

III. VISUALISATIONS

A. Visualisation 1-The comparison between China’s mental health resources and the world level

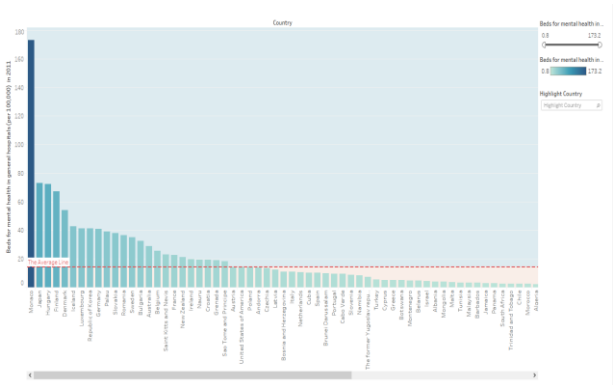


Figure 1 Beds for mental health in general hospitals (per 100,000) globally in 2011

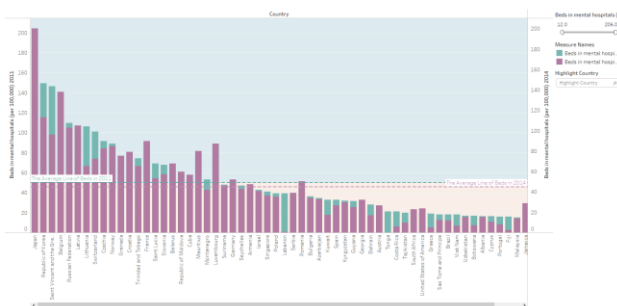


Figure 2 Beds in mental hospitals (per 100,000) around globally in 2011 and 2014

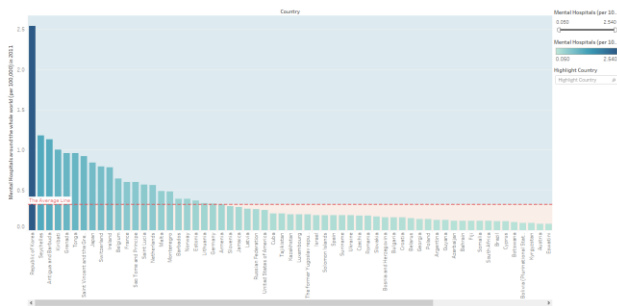


Figure 3 Mental hospitals (per 100,000) globally in 2011

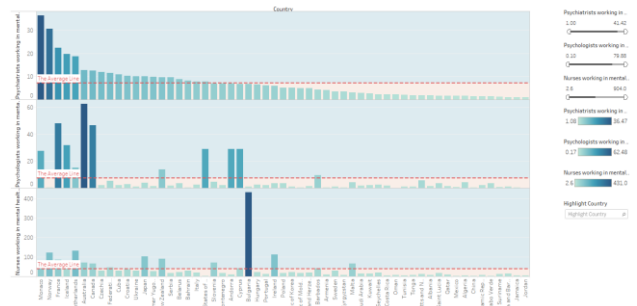


Figure 4 Mental health professionals (per 100,000) around globally in 2011

1) Description

The above bar charts illustrate the contents in order as follows:

- Psychiatric beds in general hospitals for mental health globally in 2011.
- Psychiatric beds in mental hospitals globally in 2011 and 2014.
- Mental hospitals globally in 2011.
- Psychiatrists, psychologists and nurses globally in 2011.

The interactions utilised are data range filter and highlight filter. Because the number of countries is a little more. The data range filter is used for filtering some countries to present the result audiences intend to see, especially, the purpose of this data story is to show the china’s situation, using this filter can filter the countries whose data are quite low or less than China’s. The highlight filter is utilised for highlighting some country which audiences want to see.

2) Justification

According to Toulmin model of Argumentation [8][9],

- Claims
The claim of the data story is on-line psychological services start-ups have a big potential market in China.

- Backing
The backing is China's mental health resources are below the world average level and unevenly distributed, which means China has a big demand for psychological services but lacks mental health resources.

- Warrant
The warrant of mental health spending per capita increases with Gross National Income rising. With the improvement of China’s Gross National Income, online psychological services start-ups will have a big potential market in China.

To state the backing, it is suitable to pick the relevant standard mental health resources indication data including psychiatric beds density distribution, outpatient facilities distribution and medical professionals’ distribution from the authoritative organisation-WHO.

According to Cleveland McGill principle [10][11][12][13], bar charts have the more accuracy of interpretation than pie charts and line charts. Especially for comparison, it is proper to use bar charts. According to [14] [15], the

human's first perception is the whole overview. Hence, the visualisation sorts the relevant data descending and uses the average line to indicate the purpose of this visualisation, which provides the intended audiences with an intuitional impact. Furthermore, the different background colours denoting the level of above the average line and below the average line reinforced the effect.

According to the colour principle to avoid the bias and confusion of cultural connotations [16], the colour of the data uses blue gradients in accordance with the world's medical standard indicator colour. In addition, using colour gradients to encode the changes of the same variable showing the trends intuitionally. Sanocki and Sulman's experiment pointed out the harmonious colour is helpful for a human to remember colour patterns better [17]. The bar charts of visualisation 1 use the same colour to present. Moreover, the background colour of above the average line-level uses light blue for the harmony, because the condition above the average line level is not intended to highlight. Oppositely, the condition below the average line level is designed to highlight, as a result, use the contrast colour to present the differences and distinguishing. To emphasises the key data the average line uses the high contrast colour-red colour as an accent colour to highlight. Furthermore, to highlight China, utilising the data range filter to filter the countries whose data are quite low and less than China's. The highlight filter is designed for audiences to highlight the country they intend to observe, especially for China.

In order to avoid distortion, it should avoid using dual axes, yet Figure 2 uses it, the reason for using dual axes is the data belongs to the same variable and just for two different years, the purpose is to show the trend in the two different years, it uses two high contrast colour to express the trend and avoid the confusion.

3) Narrative Design Patterns

The visualisation intends to provide highlight contrast and difference between China and other countries. As a result, the narrative design pattern of this visual is typical comparison [18]. It is used by bar charts to indicate the comparison directly in several aspects.

4) Strengths and Weaknesses

- **Strengths**
Rogers et al. propose one classification of interaction: instruction, conversation, manipulation, exploration [19]. The data range filter and highlight filter are the strengths of this visualisation. They are the first classification-instruction which allows users to make the figure do something. The second strength is the colour harmony, distinguishing and highlight.
- **Weaknesses**
The weakness is the data age is not close enough to recent years [20]. Secondly, it does not summarise and refine the data furtherly to present more concise visualisation.

5) Improvements

As mentioned at the beginning, the data of mental health resources is the most fundamental resources for severe psychiatric treatment. The topic of this data story is psychological services, it is better to find the data in psychological problems to demonstrate the topic more directly and convincing. Secondly, it will be better to summarise the data and depict the comparison relationship between China and other typical countries in one graph. The final improvement point plots the line chart showing the changing trend of China's mental health resources.

B. Visualisation 2-Are China's psychiatric beds unevenly distributed?



Figure 5 China's psychiatric bed's distribution in 2010 (per 100,000)

1) Description

The visualisation is the choropleth map to show the distribution of psychiatric beds of hospitals in different provinces of China in 2010.

The interactions designed in the visualisation are the provinces filter and highlighter filter. The provinces filter is used for displaying the provinces audiences intend to see. As well, the highlighter filter is designed for highlighting the province audiences want to observe.

2) Justification

Because the areas are provinces, it is suitable to present the data through a choropleth map [21][22][23]. Therefore, it is obvious to utilise colour gradients to presenting the difference intuitionally. The colour gradients use cold colour to warm colour gradients in line with the common use in a choropleth map.

3) Narrative Design Patterns

The visualisation uses a question as the title, actually, it is a comparison [18] among different areas of China. Thus, compare is the most suitable narrative design pattern.

4) Strengths and Weaknesses

- **Strengths**
Tidwell defines a number of navigational techniques for interactive visualisation such as search and filter, zoom, and scroll and pan etc. [24]. This visualisation utilises search and filter as a strength.
- **Weaknesses**
The weakness is the data is not picked from recent years [20].

5) Improvements

The same as the visualisation 1, it will be more accurate and convincing to obtain the data for recent years. It will be better to combine the psychiatric beds' distribution with the population density of different provinces.

C. Visualisation 3-Are China's mental health professionals unevenly distributed?

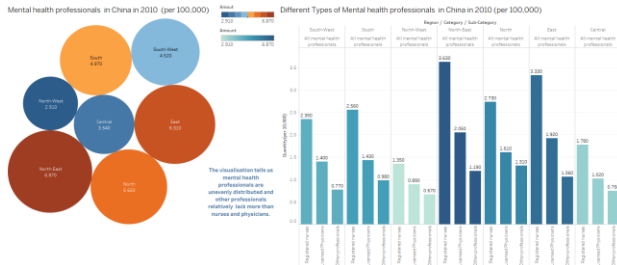


Figure 6 China's mental health professionals' distribution and types in 2010 (per 100,000)

1) Description

The visualisation uses packed bubbles and bar charts to present the distribution of mental health medical professionals in China in 2010.

There are no interactions in this visualisation because the dimension of the data is not complex.

2) Justification

Because the areas here are the comprehensive area rather than specific cities or provinces, it is not convenient to plot choropleth map as above. Therefore, it is decided to use packed bubbles graph which can visualize the size of the area for visual cues. Furthermore, the colour gradients also for visualizing the size of the area use the same gradients as the previous visualisation for colour harmony. According to the Gestalt Principles of grouping-proximity and closure [14][15], the different types of mental health professionals are categorised in one area presenting through bar charts. In addition, in terms of the number of mental health professionals, it uses the colour gradients to show the trend and the colour gradients use the same gradients as visualisation 1 for colour harmony.

3) Narrative Design Patterns

As the previous visualisation, this visualisation is another aspect of mental health resources, it is still a comparison [18].

4) Strengths and Weaknesses

- **Strengths**
This visualisation utilises one complementary graph juxtaposed [18] for supplementary explanation and presentation.
- **Weaknesses**
The weakness is the data age is not close enough to recent years [20]. In addition, packed bubbles graph is similar to pie charts, according to Cleveland McGill's theory [11], a human is not easy to distinguish the size of angle areas.

5) Improvements

The same as the visualisation 1, it will be more accurate and convincing to obtain the data for recent years.

D. Visualisation 4-Does government's mental health spending per capita increase with Gross National Income rising?

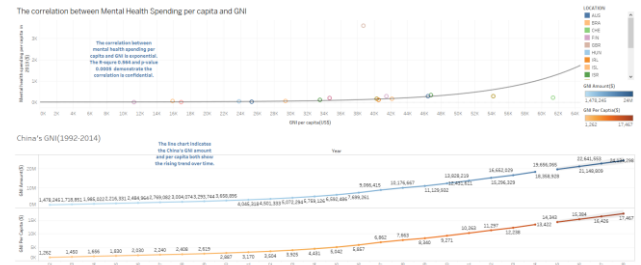


Figure 7 The correlation between Mental Health Spending per capita and GNI(\$)

1) Description

The visualisation can be divided into the two parts. The first part is to use a scatter figure to plot the map between mental health spending per capita and GNI of different WHO member states. Then, use the fit model to find the correlation between the two variables. The second part is two line charts of China's GNI amount and per capita mental health spending respectively trend from the year 1992 to 2014, and then use the forecasting model to predict the data from the year 2015 to 2018.

There are no interactions in the visualisation because the visualisation is designed for presenting the correlation and trend of corresponding variables.

2) Justification

The purpose of this visualisation is to explore the correlation between two variables, the scatter plot, therefore, is the choice then use a fitting model to fit the scatter plots. In terms of the second graph, it is suitable for using line charts to show the trend. In order to be readable, the first graph encodes different countries in different colours. The second trend line uses the colour gradients to show the changing trend and the data label for each point.

3) Narrative Design Patterns

The visualisation starts with a question [18] to provoke the engagement of audiences. The purpose is to explore the correlation between the two variables. Hence, the question pattern is more proper for this visualisation.

4) Strengths and Weaknesses

- **Strengths**
The visualisation utilises the question design pattern to provoke the engagement of audiences and make audiences explore the conclusion [18].
- **Weaknesses**
The weakness is the data is not from recent years [20].

5) Improvements

Firstly, the same as the visualisation 1, it will be more accurate and convincing to obtain the data for recent years. Secondly, according to the fitting correlation between mental health spending per capita and GNI, it can be improved by plotting an animation between China's mental health spending per capita and GNI to provide audiences with more intuitional understanding.

IV. CONCLUSION

The visualisation 1 reveals the truth that China's mental health resources are below the world average level. From visualisation 2 and 3, it can come to the conclusion that China's mental health resources are unevenly distributed. The three visualisations answered the questions 1 and 2 as desired. Also, the visualisation 4 shows the exploratory result that government's mental health spending per capita increase with Gross National Income rising, which answers question 3 as required. Overall, the visualisations accomplish the mission of this data story, to state the argument on-line psychological services start-ups have a big potential market in China.

In addition, I learned the mechanism of visual perception, varies of theories and principles of information design for the mind: Gregory's top-down processing theory and Gibson's bottom-up processing theory, the psychological principle of visual perceptual organisation, Cleveland and McGill's 10 elementary perceptual tasks, Tufte's guidelines containing the requirement of high data-ink ratio, lie factor equals 1. Especially, the most impressive gain is the colour principle: how to make colour harmonious, how to use colour to distinguish and highlight.

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